Attention Network With Fractional-Ordered Mixed Geometric Features for Hyperspectral Image Classification

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***Abstract*— In recent years, the utilization of spectral and spatial information from adjacent pixels has become one of the key focuses in hyperspectral image (HSI) classification research, and the advent of the attention mechanism has introduced innovative approaches to this classification. However, existing applications of the attention mechanism primarily concentrate on the central pixel and its similar counterparts, overlooking cases when the central pixel is a noisy outlier. To remedy this defect, we introduce an attention network with fractional-ordered mixed geometric (ANFMG) features. First, we design differential geometric attention to differentiate between noise, edge, and flat areas. When the central pixel is a noisy outlier, differential geometric attention helps the model minimize its reliance on its own spectral information and reduces the effect of surrounding noise. Second, to achieve optimal classification outcomes with limited training samples, we introduce the specific fractal detail extraction strategy for targeted extraction of detailed information and minimizing redundant input. Furthermore, to focus on the central pixel and its neighboring pixels while mitigating the influence of pixels that exhibit spectral variability, we devise a fractal and similarity measurement fusion (FSMF) module, allocating limited resources to crucial features. The efficacy of our approach is confirmed through experiments on five HSI datasets, with results indicating superior classification performance.**

***Index Terms*— Attention, curvature, fractional-order, hyperspectral image (HSI) classification.**

I. Introduction

IN RECENT years, hyperspectral images (HSIs), being a type of remote sensing data [1], have found extensive application in various domains, including urban construction [2], [3], agriculture [4], [5], [6], [7], [8], [9], marine exploration [10], [11], and disaster management [12], [13], among others. A crucial component of these applications is HSI classification, which assigns unique categories to each pixel by analyzing the spectral information of each pixel in the HSI. However, spectral variability among analogous materials, induced by atmospheric effects and instrumental factors, often results in pixel misclassification when relying solely on spectral features. In such cases, misclassified pixels are identified as noises within the image [14]. Fortunately, the majority of neighboring pixels share identical land cover labels [15], [16]. Therefore, numerous researchers opt to extract spatial features from pixel-centered 3-D blocks to alleviate this misclassification issue.

Over the past decade, employing the aforementioned strategies, convolutional neural networks (CNNs) based methods have demonstrated effectiveness in HSI classification [17], [18], [19], [20], [21], [22]. For instance, Lee and Kwon [23] proposed a contextual deep CNN that leverages local spatial–spectral relationships of pixels to extract local contextual information, thus attaining commendable classification results. Ortac and Ozcan [24] evaluated 1-D, 2-D, and 3-D convolution model methods for HSI classification. Their experimental validation confirmed efficient classification outcomes. Lu et al. [25] introduced a multi-GPU training framework (MTFC) that expedites the training of the cube-CNN model. In separate work, Li et al. [26] proposed a deep learning framework combining denoising and classification, enhancing classification performance.

However, the assumption that adjacent pixels share identical land cover labels does not hold for boundary pixels. To address the aforementioned challenges, researchers have implemented various methodologies to discern the contributions of neighboring pixels to classification [15], [16], [27], [28]. Notably, deformable convolutions and attention mechanisms have emerged as prevalent techniques. Deformable convolutions allow for the adaptation of the convolution kernel shape based on input data characteristics. This adaptability ensures a closer alignment between the convolution kernel's shape and the actual land cover, thereby enhancing the accuracy of ground target boundary classification [29], [30], [31]. Nonetheless, the utilization of a single convolutional layer proves inadequate for acquiring intricate spatial–spectral features. Consequently, recent studies have introduced the use of similarity matrices to augment deformable convolutions, enabling more precise offsets [32], [33]. In addition, the integration of superpixels with deformable convolutions has been explored to address the aforementioned challenges [34]. Regrettably, the methodologies employed in these approaches have failed to rectify misclassification stemming from central points that exhibit spectral variability. In addition to deformable convolutions, the attention mechanism stands out due to its ability to selectively focus on significant parts. It has been extensively employed in the extraction of hyperspectral-spatial and spectral features

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[14], [15], [16], [35], [36], [37], [38]. This mechanism amplifies the contribution of surrounding pixels with the same label as the center pixel to the classification while diminishing its contribution to those pixels with different labels. For example, Zheng et al. [39] proposed a rectified spatial attention (RSpaA) module for extracting rotation-invariant spectral–spatial features from HSI. Liu et al. [40] designed a scaled dot-product central attention (SDPCA) to extract spectral–spatial information from the center pixel and similar surrounding pixels on the HSI patch. The spectral–spatial attention module (SSAM), proposed by Pu et al. [41], comprises spectral and spatial branches, where the spectral branch suppresses redundant spectral information using channel attention, and the spatial branch uses spatial attention to explore contextual information. Li et al. [42] proposed a central vector-oriented self-similarity network (CVSSN) that utilizes Euclidean distance similarity measurement and cosine angle similarity measurement to fully explore the spatial relationships guided by central vectors. Zhang et al. [43] developed multiway attention modules in spectral and spatial branches to improve the extraction of spectral–spatial features. Yu et al. [44] introduced a feedback spatial attention module to refine spatial features. However, these studies mainly focused on the features of the central pixel and the features similar to the central pixel. They did not consider that when the central pixel itself is noisy, using this method might make the model overly focus on noise information, leading to classification errors.

To address the aforementioned issue, we propose an attention network with fractional-ordered mixed geometric (ANFMG) features. Initially, we design differential geometric attention helping the model suppress its attention to its own spectral information when the central pixel is a noisy outlier(a point exhibiting spectral variability). Following this, in response to the limited capacity of integer-ordered differential geometry in capturing detailed image information [45], [46], we introduce the specific fractal detail extraction strategy enabling the selective extraction of pertinent detailed features while minimizing redundant input. Subsequently, we develop a fractional-ordered differential geometric attention (FODGA) module to flexibly mitigate the impact of noise. Furthermore, a fractal and similarity measurement fusion (FSMF) module is devised to adjust the influence of the central pixel and its neighboring pixels on the algorithm's outcomes in a self-adaptive manner. ANFMG can allocate limited resources to more important spectral–spatial features, achieving higher classification accuracy. The contributions of this article can be distilled as follows.

1) We introduce the specific fractal detail extraction strategy for selective extraction of detailed information and reduction of redundant input.
2) We design a fractional-ordered differential geometry calculation (FODGC) module for the flexible extraction of noise and edge.
3) We employ the specific fractal detail extraction strategy, along with custom normalization functions and the custom fractional order derivative (FOD) approximation formula to develop an FODGA module for mitigating the impact of noise.
4) We devise an FSMF module to focus on the central pixel and its neighboring pixels while suppressing the influence of pixels that exhibit spectral variability, thereby allocating limited resources to crucial features.

The rest of this article is organized as follows. Section II introduces the related work. Section III details the proposed model. Section IV presents experimental research and discusses the results. Finally, Section V concludes the article and suggests future research directions.

II. Related Works

A. Attention Network for Distinguishing Contributions of Neighborhood Pixels

Currently, HSI classification utilizes attention mechanisms to discern the contributions of different pixels in the neighborhood. Some research directly adopted classic convolution and pooling methods to calculate the attention matrix [14], [15], [16], [35], [37], [38], [41], [43], [44]. However, some researchers were not satisfied with the classic convolution and pooling methods. They believed that it is crucial to use the central pixel as a benchmark to measure the contribution of its adjacent pixels in the classification process. Therefore, they used dot products [39], [40] or geometric features such as cosine, Euclidean distance, and so on [36], [42], to measure the similarity between the central pixel and the surrounding pixels, and then calculate the attention matrix. However, these studies mainly focused on the features of the central pixel and the features similar to the central pixel. They did not consider that when the central pixel itself is noisy, using this method might make the model overly focus on noise information, leading to classification errors.

Despite the aforementioned challenges, CVSSN explores the correlation between central spectral vectors and their neighboring counterparts in both the original input space and the feature space, resulting in commendable classification outcomes [42]. Initially, it first merges spectral and spatial vectors through the spectral–spatial information fusion (SSIF) module. Subsequently, it investigates the relationship between the central spectral vector and its neighboring counterparts using Euclidean distance and cosine similarity facilitated by the adaptive weight addition-based spectral vector self-similarity (AWA-SVSS) module and the Euclidean distance-based feature vector self-similarity (ED-FVSS) module. Finally, it efficiently acquires spectral space features through the channel spatial separation convolution (CSS-Conv) module and the scale information complementary convolution (SIC-Conv) module employing depth separable convolution. Our study draws inspiration from this model, adopting its framework. Building upon this foundation, we introduce the specific fractal detail extraction strategy, design a fractional-ordered geometric feature-based spectral vector similarity (FGF-SVS) module, and address the issue of misclassification stemming from central pixels that exhibit spectral variability.

B. Application of Fractional Order in the Hyperspectral Data

Fractional order provides a more detailed depiction than integer order, leading researchers to explore its use in

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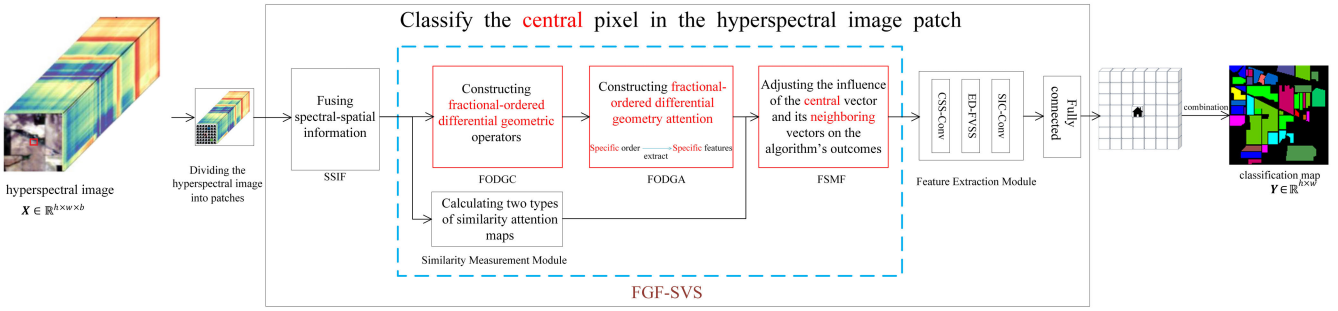


Fig. 1. Structure of the ANFMG module.

hyperspectral data. The application primarily involves two categories: utilizing FOD to extract detailed information from spectral features [47], [48], [49], and applying fractional Fourier transform (FrFT) to hyperspectral data utilization. In the second category, some researchers have used FrFT to convert the original signal from the time domain to the fractional Fourier domain, enhancing the discriminative capability between signals [50], [51]. Others have replaced self-attention layers with fractional Fourier image transformers [52], which achieve feature extraction in multiple fractional domains and reduce information loss compared to the Fourier transform. In addition, some researchers have combined Gabor and FrFT to propose fractional Gabor convolutional layers for extracting multiscale features [53]. However, these studies have only broadly extracted details from HSI for feature extraction, without selectively extracting useful features. This approach may introduce redundant information in certain cases. To precisely and efficiently extract relevant details, this study presents the specific fractal detail extraction strategy, which utilizes the custom FOD approximation formula and differential geometric measurement to extract specific details essential for HSI classification, thereby reducing redundant input.

III. Methods

The detailed structure of the proposed ANFMG is illustrated in Fig. 1. ANFMG model comprises four modules: an SSIF model, an FGF-SVS module, a feature extraction module, and a fully connected (FC) layer. Among these, FGF-SVS is an innovative module. In addition to the similarity measurement calculation (AWA-SVSS) used by previous methods, FGF-SVS adds a FODGC module, a FODGA module, and a FSMF module to solve problems that cannot be solved by similarity attention alone (highlighted in the red box). Specifically, to mitigate the limitations in representing the details of integer order and minimize redundant inputs, we utilize the specific fractal detail extraction strategy to construct the FODGC module. To address the problem of misclassification stemming from central pixels that exhibit spectral variability, we develop the FODGA and FSMF modules.

The input of ANFMG is HSI, represented as $\boldsymbol{X} \in \mathbb{R}^{h\times w\times b}$, with h, w, and b symbolizing the height, width, and quantity of spectral bands of the HSI, respectively. Each pixel, represented as $\boldsymbol{x}_{i,j} \in \mathbb{R}^{1\times 1\times b}$ in $\boldsymbol{X}$, corresponds to a semantic label $y_{i,j}$ in label set $\boldsymbol{Y} \in \mathbb{R}^{h\times w}$, where $i = 1, \ldots, h$ and $j = 1, \ldots, w$.

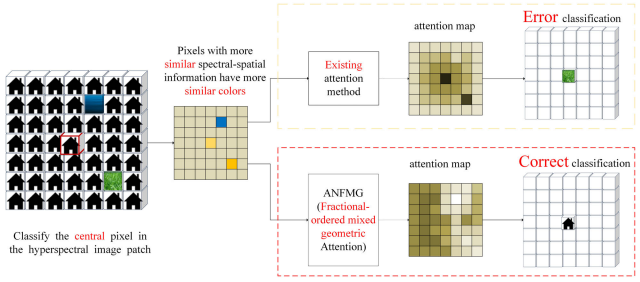


Fig. 2. Comparative analysis of ANFMG with existing methods.

The original 3-D HSI, $\boldsymbol{X}$, is divided into a set of overlapping 3-D patches, denoted as $\boldsymbol{I}$, to capitalize on the spectral and spatial information provided by the HSI. Each patch, $\boldsymbol{I}_{i,j} \in \mathbb{R}^{s\times s\times b}$ in $\boldsymbol{I}$, is a cube centered on pixel $\boldsymbol{x}_{i,j}$, with $s \times s$ indicating the area of the spatial window covered by $\boldsymbol{I}_{i,j}$. This study treats each cube, $\boldsymbol{I}_{i,j}$ in $\boldsymbol{I}$, as model input to classify the corresponding central pixel, $\boldsymbol{x}_{i,j}$, in HSI. Upon $\boldsymbol{I}_{i,j}$ input, spatial and spectral features are initially fused via SSIF. Subsequently, the FGF-SVS module utilizes self-similarity measurement, and fractional-ordered differential geometric measurement to mine the spatial relationship between the primary feature vector and its neighboring feature vectors. Then, the data proceed to CSS-Conv, ED-FVSS, and SIC-Conv to extract useful features. Eventually, the FC layer handles the concluding classification task.

ANFMG framework ensures that the model maintains its focus on pixels of the same category as the central pixel, even when the central pixel is a noise point (a point exhibiting spectral variability), as illustrated in Fig. 2.

In ANFMG, FGF-SVS is the most important module. It consists of four main parts: the FODGC module, the FODGA module, the similarity measurement module, and the FSMF module. The detailed structure of the FGF-SVS module is elucidated in Fig. 3. The input of FGF-SVS is the spectral–spatial block $\boldsymbol{I}'_{i,j} \in \mathbb{R}^{s\times s\times b'}$, an output from the SSIF module, where b' is the quantity of fused spectral channels, and $s \times s$ is the dimension of the spatial window. Subsequently, the data enters the FODGC module, FODGA module, and similarity measurement module, to generate four attention maps. Finally, the FSMF module adaptively integrates four attention maps.

In Sections III-A–III-E, we will provide a detailed introduction to the specific fractal detail extraction strategy we

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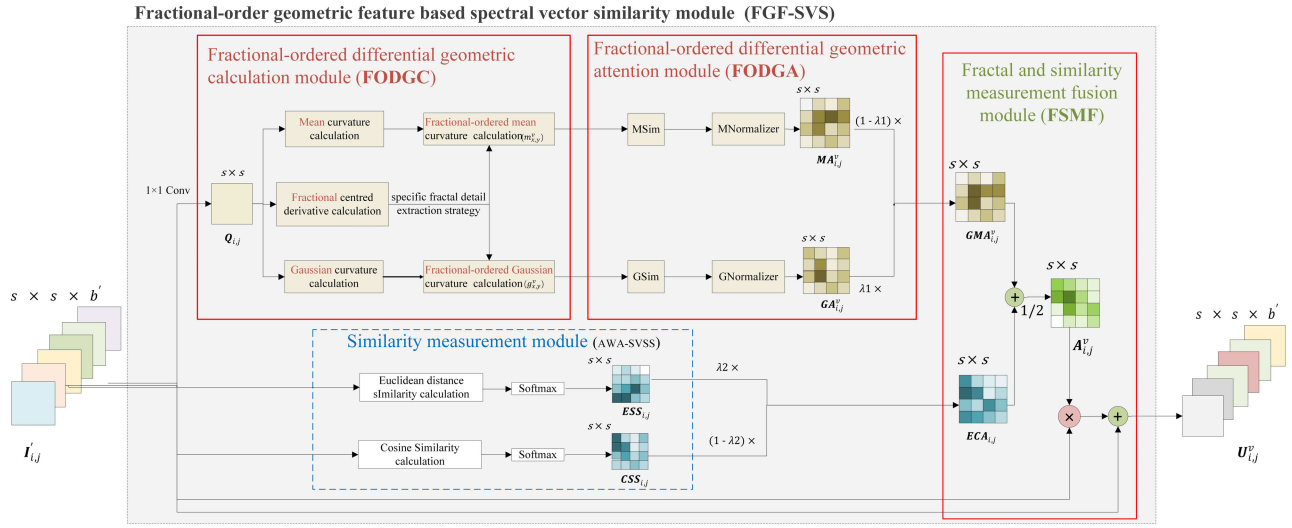


Fig. 3. Structure of the FGF-SVS module.

adopted in FGF-SVS and the four modules of the FGF-SVS: the FODGC module, the FODGA module, the similarity measurement module, and the FSMF module.

A. Specific Fractal Detail Extraction Strategy

When training data for a specific category is limited, achieving optimal performance requires inputting finer features tailored to different data. However, integer-ordered differential geometry lacks the flexibility to extract detailed features from diverse images, and the current fractional-order method applied in HSI introduces redundant information, failing to extract specific details for varying datasets. Hence, we propose the specific fractal detail extraction strategy, employing appropriate FOD and differential geometric measurement for different images to selectively extract necessary details of noise and edge. This approach overcomes the inherent limitations of integer order in representing detailed image information and adaptively retains fundamental feature details while reducing redundant information input.

In terms of the definition of FOD, there is no unified consensus in the field of mathematical theory research. Presently, the FOD used in hyperspectral data processing all employ the classical Grunwald–Letnikov (G-L) definition of the fractional integral for calculation, which is derived using backward difference. However, through analysis, we found that using the G-L definition derived from backward difference for curvature calculation will cause the coordinates of the noise obtained by the curvature to be offset. Therefore, we abandon this method in favor of using the G-L fractional centered derivative derived from central difference [54] for analysis and apply it to HSI classification. The G-L fractional centered derivative is divided into type 1 fractional centered derivative and type 2 fractional centered derivative, i.e.,

$$D^{v}_{c_1} f(t) = \lim_{h\to 0} \frac{\Gamma(v+1)}{h^v} \sum_{k\in\mathbb{Z}} \frac{(-1)^k f(t-kh)}{\Gamma\left(\frac{v+2}{2}-k\right)\Gamma\left(\frac{v+2}{2}+k\right)} \quad (1)$$

$$D^{v}_{c_2} f(t) = \lim_{h\to 0} \frac{\Gamma(v+1)}{h^v} \sum_{k\in\mathbb{Z}} \frac{(-1)^k f\left(t-kh+\frac{h}{2}\right)}{\Gamma\left(\frac{v+3}{2}-k\right)\Gamma\left(\frac{v+1}{2}+k\right)} \quad (2)$$

where $h \in \mathbb{R}^+$, Γ represents the Gamma function, and v represents the order of fractional order.

B. Fractional-Ordered Differential Geometric Calculation Module

In the FODGC module, we employ the specific fractal detail extraction strategy alongside the custom FOD approximation formula to form fractional-ordered differential geometric operators, allowing for the flexible and targeted extraction of noise and edge. By varying the orders of fractional order, fractional-ordered differential geometric operators can discern targeted features of noise and edge, thereby aiding in classification and reducing redundant input.

In the field of differential geometry, principal curvature, Gaussian curvature, and mean curvature serve as metrics reflecting the degree of a geometric object's curvature, with Gaussian curvature and mean curvature can distinguish between noise, edge, and flat areas. Therefore, we consider utilizing Gaussian curvature and mean curvature to identify noise in HSI. However, the classical Gaussian curvature and mean curvature, referring to the integer-ordered Gaussian curvature and integer-ordered mean curvature are inadequately depicted in detail and lack flexibility. Therefore, we adopt the specific fractal detail extraction strategy to construct fractional-ordered differential geometric operators: fractional-ordered Gaussian curvature $g^{v}_{x,y}$ and fractional-ordered mean curvature $m^{v}_{x,y}$, facilitating flexible extraction of noise and edge.

To compute the fractional-ordered Gaussian curvature and fractional-ordered mean curvature of the spectral–spatial block $\boldsymbol{I}'_{i,j}$, we first apply a 1×1 convolution to integrate all channels of $\boldsymbol{I}'_{i,j}$ into a single channel, thereby obtaining the fused spectral–spatial block $\boldsymbol{Q}_{i,j} \in \mathbb{R}^{s\times s}$. The $\boldsymbol{Q}_{i,j}$ can be considered a surface, which can be parameterized as $\gamma(x, y) = \{x, y, f(x, y)\}$, where $x = 1, \ldots, s$ and $y = 1, \ldots, s$, and s is the width of the spatial window. Here, x and y collectively signify the pixel's position within the spatial window, while $f(x, y)$ represents the pixel value after the fusion of the channels at the (x, y) coordinates into a single channel.

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1) $v = 1$: When the order v of fractional order is set to 1, fractional-ordered Gaussian curvature $g_{x,y}^{v}$ represents the integer-ordered Gaussian curvature $g_{x,y}$, and fractional-ordered mean curvature $m_{x,y}^{v}$ represents the integer-ordered mean curvature $m_{x,y}$.

The integer-ordered Gaussian curvature $g_{x,y}$ and integer-ordered mean curvature $m_{x,y}$ at pixel $P(x, y)$ on the surface $\boldsymbol{Q}_{i,j}$ can be calculated from the two principal curvatures [45]

$$g_{x,y} = \kappa_1 \cdot \kappa_2 \tag{3}$$

$$m_{x,y} = \frac{\kappa_1 + \kappa_2}{2} \tag{4}$$

where κ_1 and κ_2 represent the principal curvatures of the surface at nonumbilical points, perpendicular to each other.

The principal curvature κ of surface $\boldsymbol{Q}_{i,j}$ at pixel P satisfies the following equation:

$$\left(EG - F^2\right)\kappa^2 - (EN - 2FM + GL)\kappa + \left(LN - M^2\right) = 0 \tag{5}$$

where E, F, and G represent the first fundamental forms, while L, M, and N denote the second fundamental forms.

Consequently, the integer-ordered Gaussian curvature $g_{x,y}$ and integer-ordered mean curvature $g_{x,y}$ at pixel P can be determined from the first and second fundamental forms by employing the relationship between the roots of this quadratic equation and the coefficients. Their specific expressions are as follows:

$$g_{x,y} = \frac{LN - M^2}{EG - F^2} \tag{6}$$

$$m_{x,y} = \frac{EN - 2FM + GL}{2\left(EG - F^2\right)}. \tag{7}$$

Given that E, F, G, L, M, and N can be expressed in terms of the first and second derivatives of the surface $\boldsymbol{Q}_{i,j}$, the determination of the integer-ordered Gaussian curvature $g_{x,y}$ and the integer-ordered mean curvature $m_{x,y}$ at pixel P on the surface $\boldsymbol{Q}_{i,j}$ can be further defined through (8) and (9), i.e.,

$$g_{x,y} = \frac{f_{xx} f_{yy} - f_{xy}^2}{\left(1 + f_x^2 + f_y^2\right)^2} \tag{8}$$

$$m_{x,y} = \frac{f_{xx}\left(1 + f_y^2\right) - 2 f_x f_y f_{xy} + f_{yy}\left(1 + f_x^2\right)}{2\left(1 + f_x^2 + f_y^2\right)^{\frac{3}{2}}}. \tag{9}$$

At this juncture, the first derivative is obtained using central difference, i.e.,

$$f'(t) = \lim_{h \to 0} \frac{f(t + h) - f(t - h)}{2h}. \tag{10}$$

According to (3) and (4), it can be deduced that $|g_{x,y}|$ and $|m_{x,y}|$ vary in size depending on the regions of the image. They tend to be smaller in flat areas, but larger in edge and noise areas. Our analysis assumes that κ_1 is the larger principal curvature when compared with κ_2.

1) In flat areas, both κ_1 and κ_2 are small, which results in smaller values for $|g_{x,y}|$ and $|m_{x,y}|$, potentially reducing to zero.
2) In edge regions, κ_1 is large and κ_2 is small, resulting in $|g_{x,y}|$ and $|m_{x,y}|$ being larger than in flat areas.
3) In isolated noise, both κ_1 and κ_2 are large, which results in the largest values for $|g_{x,y}|$ and $|m_{x,y}|$.

2) $0 < v < 1$: When $0 < v < 1$, $g_{x,y}^{v}$ and $m_{x,y}^{v}$ are fractional-ordered Gaussian curvature and fractional-ordered mean curvature, respectively.

Utilizing (8) and (9), we compute the fractional-ordered Gaussian curvature $g_{x,y}^{v}$ and fractional-ordered mean curvature $m_{x,y}^{v}$ of the surface $\boldsymbol{Q}_{i,j}$ at pixel $P(x, y)$ using fractional-ordered first derivatives and fractional-ordered second derivatives, i.e.,

$$g_{x,y}^{v} = \frac{f_{xx}^{v} f_{yy}^{v} - f_{xy}^{v} f_{xy}^{v}}{\left(1 + f_x^{v} f_x^{v} + f_y^{v} f_y^{v}\right)^2} \tag{11}$$

$$m_{x,y}^{v} = \frac{f_{xx}^{v}\left(1 + f_y^{v} f_y^{v}\right) - 2 f_x^{v} f_y^{v} f_{xy}^{v} + f_{yy}^{v}\left(1 + f_x^{v} f_x^{v}\right)}{2\left(1 + f_x^{v} f_x^{v} + f_y^{v} f_y^{v}\right)^{\frac{3}{2}}}. \tag{12}$$

When $10v$ is even, the FOD conforms to type 1 fractional centered derivative (1). If the continuous period t of the function $f(t)$ is divided by a step size of $h = 1$, the following formula arises:

$$\begin{aligned} f^{v}(v) &= D_{c_1}^{v} f(t) \\ &\approx \Gamma(v + 1) \sum_{k \in \mathbb{Z}} \frac{(-1)^k f(t - k)}{\Gamma\left(\frac{v+2}{2} - k\right)\Gamma\left(\frac{v+2}{2} + k\right)}. \end{aligned} \tag{13}$$

For convenience in programming, this article approximates the FOD satisfying this condition by considering only the middle three terms of (13), as detailed below

$$\begin{aligned} f^{v}(v) \approx \Gamma(v + 1) \Bigg[& \frac{-1}{\Gamma\left(\frac{v}{2} + 2\right)\Gamma\left(\frac{v}{2}\right)} f(t + 1) \\ & + \frac{1}{\Gamma\left(\frac{v}{2} + 1\right)\Gamma\left(\frac{v}{2} + 1\right)} f(t) \\ & + \frac{-1}{\Gamma\left(\frac{v}{2}\right)\Gamma\left(\frac{v}{2} + 2\right)} f(t - 1) \Bigg]. \end{aligned} \tag{14}$$

When $10v$ is odd, the FOD conforms to type 2 fractional centered derivative (2). At this time, to ensure that $t - kh + h/2$ yields an integer value, the continuous period t of the function $f(t)$ is divided by a step size of $h = 2$. As a result of this partitioning, the subsequent formula is derived

$$\begin{aligned} f^{v}(v) &= D_{c_2}^{v} f(t) \\ &\approx \frac{\Gamma(v + 1)}{2^v} \sum_{k \in \mathbb{Z}} \frac{(-1)^k f(t - 2k + 1)}{\Gamma\left(\frac{v+3}{2} - k\right)\Gamma\left(\frac{v+1}{2} + k\right)}. \end{aligned} \tag{15}$$

For convenience in programming, this article approximates the FOD satisfying this condition by considering only the middle two terms of (15), as detailed below

$$\begin{aligned} f^{v}(v) \approx \frac{\Gamma(v + 1)}{2^v} \Bigg[& \frac{1}{\Gamma\left(\frac{v+3}{2}\right)\Gamma\left(\frac{v+1}{2}\right)} f(t + 1) \\ & + \frac{-1}{\Gamma\left(\frac{v+1}{2}\right)\Gamma\left(\frac{v+3}{2}\right)} f(t - 1) \Bigg]. \end{aligned} \tag{16}$$

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Through the utilization of (11) and (12), in conjunction with the custom FOD approximation formulas (14) and (16), we are able to obtain the values of $g^v_{x,y}$ and $m^v_{x,y}$. At this time, the values of $|g^v_{x,y}|$ and $|m^v_{x,y}|$ are larger in noise and edge areas, and smaller in flat areas. The orders of fractional-ordered differential geometry are selected for various datasets to extract specific details of noise and edge, enhancing flexibility, specificity, and accuracy.

C. FODGA Module

In the FODGA module, we utilize fractional-ordered differential geometric operators and custom normalization functions to construct the FODGA, distinguishing between noise, edge, and flat areas, thereby effectively mitigating noise.

In both cases of v, to achieve a greater value of fractional-ordered Gaussian curvature measurement $gm^v_{x,y}$, the pixel (x, y) becomes more crucial. $gm^v_{x,y}$ at pixel $P(x, y)$ is expressed as follows:

$$gm^v_{x,y} = -\left|g^v_{x,y}\right|. \tag{17}$$

Likewise, the fractional-ordered mean curvature measurement $mm^v_{x,y}$ at pixel $P(x, y)$ is expressed as follows:

$$mm^v_{x,y} = -\left|m^v_{x,y}\right|. \tag{18}$$

According to (17) and (18), it can be deduced that the fractional-ordered Gaussian curvature measurement $gm^v_{x,y}$ and fractional-ordered mean curvature measurement $mm^v_{x,y}$ in flat areas are larger than those in edge and noise areas.

Subsequently, the fractional-ordered Gaussian curvature measurement matrix $\boldsymbol{GM}^v_{i,j}$ and the fractional-ordered mean curvature measurement matrix $\boldsymbol{MM}^v_{i,j}$, both belonging to $\mathbb{R}^{s\times s}$, are calculated as

$$\boldsymbol{GM}^v_{i,j} = \mathrm{GSim}\left(\boldsymbol{Q}_{i,j}\right) \tag{19}$$

$$\boldsymbol{MM}^v_{i,j} = \mathrm{MSim}\left(\boldsymbol{Q}_{i,j}\right) \tag{20}$$

where GSim compute the $gm^v_{x,y}$ for each point in $\boldsymbol{Q}_{i,j}$ to determine $\boldsymbol{GM}^v_{i,j}$. Similarly, MSim calculate the $mm^v_{x,y}$ for each point in $\boldsymbol{Q}_{i,j}$ to ascertain $\boldsymbol{MM}^v_{i,j}$.

In addition, the classical softmax function for generating attention maps is not suitable here due to the exceedingly diminutive values of the negative fractional-ordered mean curvature and fractional-ordered Gaussian curvature. Using softmax would result in minimal distinction between values. Instead, custom functions, namely, MNormalizer and GNormalizer, are employed for the normalization and generation of the fractional-ordered Gaussian curvature attention map $\boldsymbol{GA}^v_{i,j} \in \mathbb{R}^{s\times s}$ and fractional-ordered mean curvature attention map $\boldsymbol{MA}^v_{i,j} \in \mathbb{R}^{s\times s}$, as detailed below

$$\boldsymbol{GA}^v_{i,j} = \mathrm{GNormalizer}\left(\boldsymbol{GM}^v_{i,j}\right) \tag{21}$$

$$\boldsymbol{MA}^v_{i,j} = \mathrm{MNormalizer}\left(\boldsymbol{MM}^v_{i,j}\right) \tag{22}$$

$$ga^v_{x,y} = \begin{cases} \dfrac{1}{s\times s}, & \text{if } gn = 0 \\ \dfrac{gm^v_{x,y} + |gn|}{\sum_{x=1}^{s}\sum_{y=1}^{s}\left(gm^v_{x,y} + |gn|\right)}, & \text{Otherwise} \end{cases} \tag{23}$$

$$ma^v_{x,y} = \begin{cases} \dfrac{1}{s\times s}, & \text{if } mn = 0 \\ \dfrac{mm^v_{x,y} + |mn|}{\sum_{x=1}^{s}\sum_{y=1}^{s}\left(mm^v_{x,y} + |mn|\right)}, & \text{Otherwise} \end{cases} \tag{24}$$

where $gn = \min(\boldsymbol{GM}^v_{i,j})$, $mn = \min(\boldsymbol{MM}^v_{i,j})$. The GNormalizer computes $\boldsymbol{GA}^v_{i,j}$ by calculating $ga^v_{x,y}$ for each pixel in $\boldsymbol{GM}^v_{i,j}$. The GNormalizer computes $\boldsymbol{MA}^v_{i,j}$ by calculating $ma^v_{x,y}$ for each pixel in $\boldsymbol{MM}^v_{i,j}$. These functions guarantee that all element values of the target attention map do not exceed one and their sum equals one.

D. Similarity Measurement Module

Euclidean distance similarity measurement and cosine angle similarity measurement serve as two effective similarity measurements. The former assesses similarity by computing the Euclidean distance between surrounding pixels and the center pixel, while the latter evaluates similarity by calculating the cosine angle between two pixels.

Specifically, the Euclidean distance similarity matrix $\boldsymbol{E}_{i,j} \in \mathbb{R}^{s\times s}$ and the Euclidean distance similarity value $e_{x,y}$ at coordinates (x, y) are computed as

$$\boldsymbol{E}_{i,j} = \mathrm{EDSim}\left(\boldsymbol{x}^0_{i,j}, \boldsymbol{I}'_{i,j}\right) \tag{25}$$

$$e_{x,y} = \mathrm{EDSimV}\left(\boldsymbol{x}^0_{i,j}, \boldsymbol{x}_{x,y}\right) = \frac{1}{1 + \left\|\boldsymbol{x}^0_{i,j} - \boldsymbol{x}_{x,y}\right\|}. \tag{26}$$

The central spectral vector $\boldsymbol{x}^0_{i,j} \in \mathbb{R}^{b'}$ is derived directly from $\boldsymbol{I}'_{i,j}$. EDSim calculates $\boldsymbol{E}_{i,j}$ using EDSimV, which determines the Euclidean similarity between the central pixel $\boldsymbol{x}^0_{i,j}$ and each neighboring spectral vector $\boldsymbol{x}_{x,y} \in \mathbb{R}^{b'}$ within the spectral space patch $\boldsymbol{I}'_{i,j}$. A higher similarity suggests a value closer to one.

Similarly, the cosine similarity matrix $\boldsymbol{C}_{i,j} \in \mathbb{R}^{s\times s}$ and the cosine similarity value $c_{x,y}$ at coordinates (x, y) are determined as follows:

$$\boldsymbol{C}_{i,j} = \mathrm{CosSim}\left(\boldsymbol{x}^0_{i,j}, \boldsymbol{I}'_{i,j}\right) \tag{27}$$

$$c_{x,y} = \mathrm{CosSimV}\left(\boldsymbol{x}^0_{i,j}, \boldsymbol{x}_{x,y}\right) = \frac{\boldsymbol{x}^0_{i,j}\cdot\boldsymbol{x}_{x,y}}{\left\|\boldsymbol{x}^0_{i,j}\right\|\left\|\boldsymbol{x}_{x,y}\right\|} \tag{28}$$

where CosSim computes $C_{i,j}$ via CosSimV, which assesses the cosine angle similarity between $x^0_{i,j}$ and each $x_{x,y}$. A greater similarity indicates a value closer to one.

Moreover, we apply the softmax function to normalize and generate the Euclidean distance self-similarity attention map, $\boldsymbol{ESS}_{i,j} \in \mathbb{R}^{s\times s}$, and cosine self-similarity attention map, $\boldsymbol{CSS}_{i,j} \in \mathbb{R}^{s\times s}$. In detail

$$\boldsymbol{ESS}_{i,j} = \mathrm{Softmax}\left(\boldsymbol{E}_{i,j}\right) \tag{29}$$

$$\boldsymbol{CSS}_{i,j} = \mathrm{Softmax}\left(\boldsymbol{C}_{i,j}\right). \tag{30}$$

E. FSMF Module

In the FSMF module, we adaptively integrate four distinct attention maps to focus on the central pixel and its neighboring pixels while mitigating the influence of pixels that exhibit

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spectral variability, thereby allocating limited resources to crucial features.

Based on the preceding analysis, it is evident that higher values in matrices $\boldsymbol{ESS}_{i,j}$, $\boldsymbol{CSS}_{i,j}$, $\boldsymbol{GA}^{v}_{i,j}$, and $\boldsymbol{MA}^{v}_{i,j}$ correlate with increased significance in classification. Consequently, we adopt an adaptive weighting addition method to ensure that different attention maps do not interfere with each other but rather adaptively combine based on the model and the data itself.

Initially, we partition variable fractal measurements into one group and similarity measurements into another group. The groups will be integrated using the adaptive weighting addition method, as detailed below

$$\boldsymbol{GMA}^{v}_{i,j} = \lambda 1 \times \boldsymbol{GA}^{v}_{i,j} + (1-\lambda 1) \times \boldsymbol{MA}^{v}_{i,j} \tag{31}$$

$$\boldsymbol{ECA}_{i,j} = \lambda 2 \times \boldsymbol{ESS}_{i,j} + (1-\lambda 2) \times \boldsymbol{CSS}_{i,j} \tag{32}$$

where $\boldsymbol{GMA}^{v}_{i,j}$ is an adaptively FODGA map, created by combining $\boldsymbol{GA}^{v}_{i,j}$ and $\boldsymbol{MA}^{v}_{i,j}$ via an adaptive weighting addition method. The fractional-ordered Gaussian curvature cannot differentiate between edge and flat areas. When the network requires edge information, adaptively decreasing the value of $\lambda 1$ enhances the contribution of the fractional-ordered mean curvature, whereas increasing $\lambda 1$ enhances the contribution of the fractional-ordered Gaussian curvature. Similarly, $\boldsymbol{ECA}_{i,j}$ is generated by amalgamating the Euclidean distance self-similarity attention map $\boldsymbol{ESS}_{i,j}$ with the cosine self-similarity attention map $\boldsymbol{CSS}_{i,j}$, utilizing the same adaptive weighting addition approach. Both $\lambda 1$ and $\lambda 2$ commence with an initial value of 0.5, being subject to adaptive optimization throughout the training phase of the model.

Moreover, the fractional-ordered mixed geometric attention map $\boldsymbol{A}^{v}_{i,j} \in \mathbb{R}^{s\times s}$ is calculated as follows:

$$\boldsymbol{A}^{v}_{i,j} = \left(\boldsymbol{GMA}^{v}_{i,j} + \boldsymbol{ECA}_{i,j}\right)/2. \tag{33}$$

To ascertain that every value of $\boldsymbol{A}^{v}_{i,j}$ remains below one, we implement a division by two.

At this stage, $\boldsymbol{GMA}^{v}_{i,j}$ values are smaller in noise and edge areas but larger in flat areas. When $\boldsymbol{ECA}_{i,j}$ remains constant, the $\boldsymbol{A}^{v}_{i,j}$ value follows the same trend. Thus, when the central pixel is a noisy outlier, $\boldsymbol{GMA}^{v}_{i,j}$ reduces the contribution of the central pixel, prompting the model to learn more from the surrounding pixels in flat areas rather than it is own features. When the central pixel is surrounded by flat areas, $\boldsymbol{A}^{v}_{i,j}$ is dominated by $\boldsymbol{ECA}_{i,j}$, and $\boldsymbol{GMA}^{v}_{i,j}$ remains inactive.

The final feature map $\boldsymbol{U}^{v}_{i,j} \in \mathbb{R}^{s\times s\times b'}$, is generated as

$$\boldsymbol{U}^{v}_{i,j} = \boldsymbol{A}^{v}_{i,j} \bigotimes \boldsymbol{I}'_{i,j} + \boldsymbol{I}'_{i,j} \tag{34}$$

where $\bigotimes$ denotes element-wise multiplication. During the multiplication process, attention values are broadcast correspondingly. Notably, an element-wise addition operation with $\boldsymbol{I}'_{i,j}$ is executed to prevent vanishing gradient and network degradation.

IV. Experiment

A. Datasets Description

In this study, the effectiveness of the proposed ANFMG model is validated by conducting experiments on five widely-used hyperspectral datasets: Indian Pines (IP), University of Pavia (UP), Kennedy Space Center (KSC), University of Houston (UH), and WHU-Hi-LongKou (LK). Among them, the IP data, originating from a region of IP in Indiana and imaged by AVIRIS in 1992, was the inaugural test data for HSI classification. At the same time, the Houston2013 dataset, encompassing imagery from the UH campus and adjacent urban areas, boasts a resolution of 349 × 1905 pixels. Meanwhile, the UP data, collected by the ROSIS-03 in central Pavia, Italy, consists of 103 effective bands, nine types of ground information, and dimensions of 610 × 340. Furthermore, the KSC data, collected by the AVIRIS sensor at the KSC in Florida, comprises 224 bands, of which 176 remain after water vapor noise removal. The spatial resolution is 18 m with dimensions of 512 × 614 pixels, and it contains 13 categories. Finally, the WHU-Hi-LongKou dataset was gathered in Longkou Town, Hubei Province, China, in 2018. It comprises images with dimensions of 550 × 400 pixels, containing 270 bands, and encompassing nine categories. Basic information for these datasets is tabulated in Table I.

In terms of sample selection, this study explores two partitioning methods. First, for each dataset, 10% and 1% of the total labeled pixels per category are randomly selected as the training and validation sets, respectively, while the remaining labeled pixels form the test set. Second, a partitioning approach based on spatially disjoint samples is adopted. The spatially disjoint training and testing samples for the UP dataset are accessible on the GRSS DASE website, while those for the LK and UH datasets are provided by the authorities. In this article, each class in the LK training set contains 100 samples. Except for experiments in Section IV-H, the first partition method is used for all other experiments.

B. Experimental Settings and Evaluation Metrics

The experiments are conducted on a workstation equipped with an Intel Core CPU i9-10920X, 64 GB RAM, and a single Nvidia GeForce RTX 3090 GPU with 24 GB GPU memory. The system operates on Ubuntu 20.04.4 LTS 64-bit, and the proposed ANFMG algorithm is implemented using Python-3.8.10 with the PyTorch-1.8.1 framework. For the training of ANFMG, the Adam optimizer is employed. The batch size, learning rate, and number of training epochs are set at 32, 0.001, and 100, respectively. Moreover, before cropping 3-D blocks for each HSI pixel, a zero-padding operation is performed to fill $(s-1)/2$ zero pixels from each HSI pixel. Furthermore, the spatial window size is set to 9.

To systematically evaluate and compare the classification performance of various methods, four common evaluation metrics are adopted: overall accuracy (OA), average accuracy (AA), kappa coefficient (Kappa), and accuracy of each class (AEC).

C. Determining the Orders of Fractional Order

The degree of attention to image detail inversely corresponds with the orders of fractional order, indicating that smaller orders yield higher attention to detail and different images require specific order settings. As such, it is essential

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TABLE I
NUMBER OF SAMPLES FOR IP, KSC, UP, UH, AND LK DATASETS

Category	IP		K		UP		UH		LK	
	Class name	Number of samples	Class name	Number of samples	Class name	Number of samples	Class name	Number of samples	Class name	Number of samples
1	Alfalfa	46	Scrub	761	Asphalt	6631	Healthy grass	1251	Corn	34511
2	Corn-notill	1428	Willow-swamp	243	Meadows	18649	Stressed grass	1254	Cotton	8374
3	Corn-mintill	830	Cp-hammock	256	Gravel	2099	Synthetic grass	697	Sesame	3031
4	Corn	237	Cp/Oak	252	Trees	3064	Trees	1244	Broad-leaf-soybean	63212
5	Grass-pasture	483	Slash-pine	161	Painted metal sheets	1345	Soil	1242	Narrow-leaf-soybean	4151
6	Grass-trees	730	Oak/Broadleaf	229	Bare Soil	5029	Water	325	Rice	11854
7	Grass-pasture-mowed	28	Hardwood-swamp	105	Bitumen	1330	Residential	1268	Water	67056
8	Hay-windrowed	478	Graminoid-marsh	431	Self-Blocking Bricks	3682	Commercial	1244	Roads and houses	7124
9	Oats	20	Spartina-marsh	520	Shadows	947	Road	1252	Mixed weed	5229
10	Soybean-notill	972	Cattail-marsh	404			Highway	1227		
11	Soybean-mintill	2455	Salt-marsh	419			Railway	1235		
12	Soybean-clean	593	Mud-flats	503			Parking Lot 1	1233		
13	Wheat	205	Water	927			Parking Lot 2	469		
14	Woods	1265					Tennis Court	428		
15	Buildings-Grass-Trees-Drives	386					Running Track	660		
16	Stone-Steel-Towers	93								
	Total	10249	Totel	5211	Total	42776	Total	15029	Total	204542

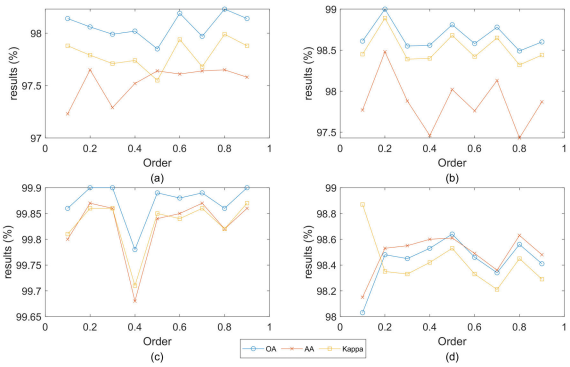


Fig. 4. Comparison of ANFMG experimental results under different order conditions in different datasets. (a) IP dataset. (b) KSC dataset. (c) UP dataset. (d) UH dataset.

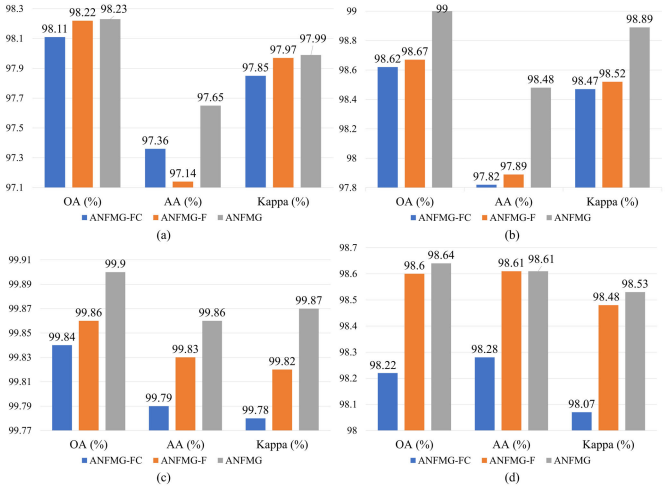


Fig. 5. Comparison of classification results of ANFMG and its variants across datasets (%). (a) IP dataset. (b) KSC dataset. (c) UP dataset. (d) UH dataset.

to tune this parameter according to each unique dataset. In the experiment, we selected nine different orders commonly used in experiments such as 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, and 0.9 for experimental analysis, with the results shown in Fig. 4. From this, we found the ideal orders for IP, KSC, UP, and UH to be 0.8, 0.2, 0.9, and 0.5, respectively. On the UH dataset, the Kappa achieves maximum value when the order equals 0.1, but both OA and AA exhibit relatively low values. After thorough deliberation, we opt for 0.5 on the UH dataset. On the UP dataset, minimal distinctions exist among orders of 0.2, 0.3, and 0.9. In this study, we opt for 0.9.

D. Ablation Experiment

To delve deeper into the proposed differential geometric measurement and FOD contributions, we performed ablation experiments on four datasets: IP, KSC, UP, and UH. Specifically, we compared our proposed ANFMG model with the following variants of ANFMG.

1) *ANFMG-F:* This version of ANFMG replaces the FOD in ANFMG with an integer order derivative.
2) *ANFMG-FC:* This variant removes the differential geometric measurement from ANFMG-F, aligning its network structure with the CVSSN model.

The performance of these models was quantitatively assessed and the results are illustrated in Fig. 5.

Compared to ANFMG, ANFMG-F shows varying degrees of decline in OA, AA, and Kappa coefficients across the datasets, indicating the FOD helps the model to extract details that are effective for classification, and efficiently find useful features in small-scale data. Particularly, the KSC dataset has the most significant decline, with OA dropping by 0.33%, AA by 0.59%, and Kappa by 0.37%

Compared to ANFMG-F, ANFMG-FC experienced a decline in most datasets, with the exception of an increase in the AA value for the IP dataset. The UH dataset exhibited the most significant drops with a 0.38% decrease in OA, a 0.41% decrease in the Kappa coefficient, and a 0.33% decrease in AA. These changes across various datasets indicate that the differential geometric measurement has the capability to extract relevant information from the surroundings and filter out noncontributing data such as noise. Further analysis revealed that the AA value for the IP dataset did not meet expectations due to the insufficient sample size in the Grass-pasture-mowed category. To ameliorate this, FOD was introduced into the model to discern finer details. Experimental results demonstrated that with this refinement, the AA value for the IP dataset saw a notable improvement.

In conclusion, the differential geometric measurement effectively suppresses irrelevant information such as noise, while the FOD helps the model to extract critical details for classification. Therefore, it is evident that the integration of differential geometric measurement and FOD is indispensable in the ANFMG algorithm.

E. Comparative Experiment

To assess the validity of ANFMG, the first partitioning method is employed to split the IP, KSC, UP, and UH datasets

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TABLE II
CLASSIFICATION PERFORMANCE OF DIFFERENT ALGORITHMS IN IP DATASET (%)

Class	SVM [55]	RF [56]	ContextNet [23]	RSSAN [16]	SSTN [37]	SSAN [15]	SSSAN [36]	SSAtt [14]	A^2S^2K-ResNet [35]	CVSSN [42]	ANFMG (our)
1	47.09±13.90	71.48±22.18	93.73±7.58	93.46±15.20	92.24±7.74	92.39±13.11	**96.59±5.01**	94.94±10.75	91.60±11.32	96.49±4.08	93.49±8.61
2	75.00±1.88	68.75±1.87	78.50±3.09	82.38±4.85	92.11±6.70	87.84±2.04	92.53±2.25	93.20±1.44	**98.21±1.18**	97.57±0.80	97.68±0.77
3	76.98±4.34	75.94±2.92	81.20±4.95	84.83±6.08	94.95±4.52	87.28±2.75	96.63±2.02	96.49±1.35	97.63±2.26	98.90±0.95	**98.99±1.06**
4	63.31±4.37	57.29±5.23	85.04±6.50	85.02±4.99	96.29±4.51	88.94±6.93	94.80±4.22	97.42±2.52	96.20±3.54	98.06±1.82	**98.87±1.43**
5	89.72±3.01	85.67±3.21	94.21±2.39	94.05±1.37	92.95±3.66	94.49±4.24	97.77±1.64	97.82±1.40	98.25±1.78	98.12±1.41	**98.89±0.70**
6	89.13±1.56	81.33±1.52	95.07±2.20	92.73±2.58	97.49±2.63	93.94±3.05	98.26±2.07	98.70±1.25	**99.86±0.21**	99.22±0.78	99.34±0.80
7	85.52±8.07	32.95±42.12	91.84±8.07	82.97±12.36	75.89±15.04	86.45±12.41	90.54±5.97	**97.44±4.75**	90.08±10.55	94.82±10.70	94.53±6.61
8	91.29±1.50	85.14±1.72	95.75±2.39	94.49±3.20	99.39±0.61	95.87±2.80	98.03±1.64	98.89±0.80	98.71±0.79	99.52±0.70	**99.65±0.64**
9	65.94±13.55	53.33±44.44	92.57±12.85	87.71±11.31	77.72±13.22	83.97±16.40	89.79±9.03	91.84±13.32	87.46±13.29	92.67±8.62	**96.13±5.41**
10	74.12±2.03	72.80±4.36	87.50±3.33	86.52±5.32	90.99±5.89	91.07±1.73	94.58±1.94	95.39±2.16	97.04±1.76	**97.66±1.41**	97.46±1.39
11	75.57±1.33	69.56±1.49	87.89±1.87	87.45±3.55	95.88±2.40	91.14±1.40	96.31±1.16	95.50±1.55	98.16±0.64	**98.30±0.64**	98.24±0.97
12	78.09±4.79	62.50±5.74	75.05±4.34	72.98±4.42	93.48±4.98	83.64±2.56	87.03±4.34	92.35±2.32	96.27±2.34	97.04±1.18	**97.68±1.11**
13	91.67±5.28	85.43±4.80	97.04±3.71	93.69±4.62	97.92±3.33	96.96±2.96	**99.62±0.64**	99.16±1.54	**99.62±0.84**	99.46±0.72	99.51±0.66
14	91.43±1.21	89.64±0.94	96.31±1.28	94.32±1.92	97.14±1.60	96.70±1.51	98.23±0.81	97.51±1.33	98.79±0.92	98.56±0.67	**98.91±0.60**
15	78.52±5.21	63.53±5.85	88.97±3.52	84.85±5.06	91.01±4.05	91.03±2.96	93.91±1.72	94.36±2.53	95.22±1.64	**95.72±1.91**	94.95±1.48
16	97.44±2.67	97.12±2.56	93.70±5.23	88.86±6.70	92.92±6.71	91.23±9.50	97.12±3.15	93.58±5.67	96.97±2.39	95.61±2.29	**98.02±2.24**
OA (%)	80.31±0.49	75.10±0.83	87.71±1.19	87.54±2.68	94.46±2.65	91.25±1.25	95.51±0.86	95.81±0.53	97.89±0.29	98.11 ± 0.15	**98.23±0.42**
AA (%)	79.43±1.69	72.03±3.91	89.65±1.42	87.90±2.20	92.40±1.68	90.81±1.83	95.11±1.04	95.91±1.19	96.25±1.46	97.36 ± 0.68	**97.65±0.91**
Kappa (%)	77.40±0.59	71.26±0.95	85.93±1.36	85.74±3.09	93.68±3.02	90.00±1.43	94.87±0.99	95.22±0.61	97.59±0.33	97.85 ± 0.17	**97.99±0.48**

TABLE III
CLASSIFICATION PERFORMANCE OF DIFFERENT ALGORITHMS IN KSC DATASET (%)

Class	SVM [55]	RF [56]	ContextNet [23]	RSSAN [16]	SSTN [37]	SSAN [15]	SSSAN [36]	SSAtt [14]	A^2S^2K-ResNet [35]	CVSSN [42]	ANFMG (our)
1	91.15±1.31	91.00±1.35	98.70±1.19	98.18±1.53	99.74±0.50	98.69±1.25	99.10±0.96	99.50±0.54	99.07±0.80	99.82±0.39	**99.88±0.27**
2	89.27±3.73	77.73±5.88	81.77±9.20	92.01±6.64	93.03±3.89	87.37±11.78	95.05±2.53	96.99±2.02	96.50±2.25	**98.45±1.21**	98.37±1.68
3	69.73±5.62	87.72±4.79	77.01±4.70	81.83±4.07	90.91±6.97	75.00±4.66	85.04±11.02	88.74±2.68	89.15±4.25	**97.82±2.14**	96.81±2.53
4	48.94±3.33	59.53±4.27	69.95±5.31	71.66±5.58	79.94±8.38	74.06±5.85	81.59±8.46	83.23±5.92	86.44±7.23	91.73±8.20	**93.23±3.96**
5	70.71±10.19	71.04±6.34	64.56±4.76	71.11±11.95	74.14±12.85	69.69±10.15	83.72±8.15	83.75±6.78	88.27±6.91	92.70±7.10	**97.50±2.69**
6	71.96±7.49	61.91±7.15	88.90±4.36	84.49±6.08	95.65±3.33	84.83±5.65	90.90±6.58	91.86±5.90	92.30±5.98	94.39±4.17	**97.55±2.41**
7	75.95±4.12	73.51±3.32	88.28±5.72	94.29±3.71	92.31±3.87	83.03±7.64	93.25±5.36	98.09±3.88	96.65±2.88	98.76±1.44	**98.92±2.63**
8	88.34±3.78	76.36±4.66	92.40±4.28	97.12±1.97	96.97±3.00	93.00±5.74	97.27±2.75	98.26±1.25	97.28±2.20	99.28±0.76	**99.29±1.19**
9	89.19±2.66	84.62±2.64	95.51±2.84	97.89±1.49	98.46±1.12	95.43±3.25	99.20±0.99	99.17±0.92	99.15±0.92	99.59±0.52	**99.76±0.44**
10	97.22±2.91	89.14±6.54	93.72±4.14	97.05±2.48	98.47±1.60	92.93±5.44	99.72±0.68	99.83±0.25	99.51±0.91	**99.97±0.08**	99.76±0.73
11	96.51±1.18	97.57±1.95	98.91±1.59	99.48±0.74	99.47±0.94	99.79±0.38	**99.92±0.17**	99.55±0.77	99.24±1.00	99.21±2.29	99.66±0.77
12	93.40±2.32	92.08±2.69	95.65±3.90	95.37±4.22	98.12±1.95	96.00±2.35	98.89±1.09	99.29±0.68	98.77±1.38	**99.87±0.23**	99.60±0.60
13	99.81±0.21	98.39±0.63	99.61±0.66	99.93±0.10	99.56±0.77	99.39±0.58	99.95±0.14	99.76±0.64	**100.00±0.00**	**100.00±0.00**	99.90±0.29
OA (%)	88.20±0.42	86.38±0.97	92.48±1.24	94.18±1.46	96.07±0.75	92.63±2.11	96.38±1.30	97.09±0.48	97.14±0.64	98.62 ± 0.54	**99.00±0.32**
AA (%)	83.24±1.26	81.59±1.52	88.08±1.76	90.80±2.14	93.60±1.45	88.40±3.13	94.12±1.80	95.23±0.89	95.56±1.03	97.82 ± 0.82	**98.48±0.56**
Kappa (%)	86.84±0.47	84.82±1.08	91.62±1.38	93.52±1.62	95.63±0.84	91.79±2.34	95.97±1.45	96.77±0.54	96.81±0.72	98.47 ± 0.60	**98.89±0.36**

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT ALGORITHMS IN UP DATASET (%)

Class	SVM [55]	RF [56]	ContextNet [23]	RSSAN [16]	SSTN [37]	SSAN [15]	SSSAN [36]	SSAtt [14]	A^2S^2K-ResNet [35]	CVSSN [42]	ANFMG (our)
1	93.89±0.42	91.61±0.99	98.83±0.31	98.22±0.46	99.01±0.74	99.18±0.31	99.73±0.16	99.34±0.29	99.86±0.15	99.85±0.10	**99.91±0.10**
2	94.58±0.25	89.76±0.32	99.76±0.11	99.84±0.06	99.67±0.26	99.85±0.05	99.88±0.07	99.91±0.04	99.95±0.04	99.94±0.06	**99.97±0.04**
3	85.62±0.99	79.48±1.81	96.98±1.12	97.85±0.71	99.39±0.76	98.73±0.80	99.49±0.43	99.51±0.39	99.50±0.50	99.77±0.27	**99.97±0.08**
4	97.08±0.49	93.43±0.81	99.64±0.23	99.49±0.29	97.71±1.84	99.76±0.13	99.77±0.14	99.83±0.11	**99.92±0.10**	99.78±0.21	99.80±0.26
5	99.39±0.48	98.83±0.65	99.92±0.13	99.62±0.48	99.83±0.17	99.97±0.06	**99.98±0.05**	99.93±0.12	99.89±0.16	99.87±0.19	99.91±0.14
6	94.54±0.23	90.80±0.49	99.48±0.33	99.59±0.19	99.63±0.48	99.81±0.21	99.84±0.12	99.69±0.22	**100.00±0.00**	99.91±0.10	99.90±0.13
7	88.93±1.28	84.86±1.44	98.93±0.44	98.03±0.93	97.25±5.53	99.21±0.52	99.63±0.39	99.52±0.61	99.91±0.15	99.79±0.44	**99.97±0.06**
8	83.09±0.83	80.11±0.89	96.71±0.81	96.36±0.89	95.34±3.34	98.27±0.50	98.93±0.40	98.41±0.59	99.18±0.38	99.27±0.32	**99.60±0.23**
9	99.96±0.05	99.88±0.11	99.73±0.29	99.14±0.86	98.56±0.94	99.86±0.17	99.99±0.04	99.93±0.11	**100.00±0.00**	99.89±0.22	99.70±0.61
OA (%)	93.28±0.10	89.40±0.24	99.15±0.14	99.06±0.17	98.90±0.62	99.53±0.11	99.74±0.06	99.63±0.08	99.85±0.02	99.84 ± 0.08	**99.90±0.05**
AA (%)	93.01±0.20	89.86±0.21	98.88±0.22	98.68±0.29	98.49±0.89	99.40±0.15	99.70±0.11	99.56±0.09	99.80±0.04	99.79 ± 0.12	**99.86±0.09**
Kappa (%)	91.02±0.14	85.71±0.33	98.87±0.19	98.75±0.22	98.55±0.82	99.37±0.15	99.66±0.08	99.51±0.11	99.80±0.02	99.78 ± 0.10	**99.87±0.06**

into training and testing sets, and the proposed ANFMG is compared against ten related algorithms: support vector machine (SVM), random forest (RF), ContextNet, RSSAN, SSTN, SSAN, SSSAN, SSAtt, A^2S^2K-ResNet, and CVSSN. SVM and RF are commonly used methods. ContextNet utilizes the joint multiscale spatial–spectral feature to explore local contextual information, achieving good classification results. The other nine advanced methods, like the proposed ANFMG method, are committed to discerning the contributions of various pixels within the neighborhood to classification.

The entire procedure was repeated a total of ten times to document both the mean accuracy and standard deviation, maintaining consistency among all parameters barring the variables throughout the comparative analysis.

1) Quantitative Analysis: The mean classification results and their associated standard deviations for IP, KSC, UP, and UH datasets are presented in Tables II–V. Notably, ANFMG exhibited superior performance across all four datasets. Specifically, for the IP dataset, our ANFMG model improved OA, AA, and Kappa values by 0.12%, 0.29%, and 0.14%, respectively, against the next-best method, CVSSN. Improvements of 0.38%, 0.66%, and 0.42% were noted on the KSC dataset. On the UH dataset, there were increases of 0.42%, 0.33%, and 0.46%, respectively. Across the UP dataset, our ANFMG model demonstrates enhancements of 0.05%,0.06%, and 0.07% compared to the next-best method A^2S^2K-ResNet, and improvements of 0.06%,0.07%, and 0.09% over CVSSN, respectively. Tables II–V show relatively low standard deviations, indicating good robustness of ANFMG. In terms of classwise accuracy across the four datasets, our model demonstrated the highest classification accuracy in 8 of 16 classes for the IP dataset, 7 of 13 classes for the KSC dataset, 5 of 9 classes for the UP dataset, and 5 of 15 classes for the UH dataset.

ANFMG not only utilizes spectral information, as in RF and SVM, but also utilizes spatial information. In addition,

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TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT ALGORITHMS IN UH DATASET (%)

Class	SVM [55]	RF [56]	ContextNet [23]	RSSAN [16]	SSTN [37]	SSAN [15]	SSSAN [36]	SSAtt [14]	A^2S^2K-ResNet [35]	CVSSN [42]	ANFMG (our)
1	94.40±1.00	96.93±0.81	96.93±1.23	95.82±2.23	96.01±3.95	95.90±4.71	92.66±6.06	94.42±3.88	97.99±1.23	**98.87±0.64**	98.77±0.58
2	94.84±1.72	94.94±1.68	97.08±0.82	93.94±3.88	91.76±6.23	97.96±1.65	96.04±2.63	96.82±2.23	99.36±0.48	99.65±0.23	**99.71±0.34**
3	99.60±0.26	99.55±0.39	98.57±1.56	98.41±2.78	**99.82±0.25**	99.15±0.99	99.12±0.79	97.10±5.83	99.74±0.36	99.40±0.63	99.59±0.61
4	**99.11±0.42**	98.00±1.03	97.28±1.16	96.74±3.70	95.74±4.52	98.36±0.78	98.69±0.95	97.44±1.77	96.98±2.49	96.68±3.75	99.10±0.57
5	94.30±1.63	92.17±1.06	99.49±0.32	98.30±3.55	99.53±0.77	97.93±5.41	99.68±0.37	99.74±0.27	**99.92±0.19**	99.46±0.36	99.62±0.49
6	99.34±1.17	96.51±4.09	96.84±2.34	94.26±2.65	94.95±2.35	96.04±4.60	93.76±7.85	98.22±1.31	**99.78±0.56**	97.57±2.06	96.96±3.10
7	77.76±2.31	82.02±2.43	92.23±1.43	88.97±5.72	93.42±4.25	91.43±4.85	93.55±2.39	94.07±1.11	96.96±1.68	95.95±1.09	**97.26±1.30**
8	82.37±2.84	86.22±1.80	90.58±4.16	90.80±3.94	94.02±6.39	95.61±3.92	92.71±6.29	94.37±5.20	97.73±1.54	98.97±1.17	**99.24±0.46**
9	72.03±0.97	75.47±2.55	90.28±1.55	90.17±2.11	94.02±3.36	91.17±5.43	93.55±3.58	94.38±2.30	97.65±1.24	97.57±1.18	**97.90±0.92**
10	81.10±2.65	84.23±3.37	90.89±2.60	87.43±7.29	94.79±2.42	91.08±4.62	92.02±4.54	92.16±2.95	96.97±1.60	98.14±1.44	**99.10±0.64**
11	76.88±1.41	77.36±1.51	91.76±3.12	92.99±2.84	94.67±3.39	94.46±2.02	96.60±1.88	95.42±2.01	97.92±0.85	**99.18±1.37**	98.73±0.81
12	72.23±2.49	72.42±2.32	91.29±2.30	89.86±4.32	93.82±2.00	92.79±2.98	94.18±2.45	91.62±4.84	95.62±1.79	**97.25±0.88**	96.86±1.71
13	85.99±8.64	79.35±7.04	96.36±1.96	94.62±4.09	91.77±5.58	96.57±2.81	95.79±3.46	96.62±5.32	**98.16±1.43**	96.66±2.60	97.61±2.05
14	93.60±1.21	93.58±2.80	98.07±3.91	97.78±2.24	99.02±1.85	97.68±1.30	**99.82±0.20**	98.86±1.02	99.24±2.03	99.25±1.63	99.34±1.25
15	99.05±0.71	98.24±0.60	98.64±1.40	97.72±1.38	98.44±1.06	98.69±1.56	98.37±1.56	97.94±3.09	99.09±0.74	**99.63±0.75**	99.36±0.57
OA (%)	86.15±0.86	87.26±0.40	94.40±0.88	93.02±2.10	94.98±1.49	95.07±2.83	95.26±1.39	95.35±1.16	97.94±0.42	98.22±0.49	**98.64±0.24**
AA (%)	88.17±1.10	88.47±0.83	95.09±0.88	93.85±1.71	95.45±1.06	95.65±2.22	95.77±1.40	95.95±0.97	98.21±0.36	98.28±0.33	**98.61±0.33**
Kappa (%)	85.01±0.93	86.21±0.43	93.95±0.95	92.45±2.27	94.57±1.61	94.67±3.07	94.88±1.50	94.97±1.25	97.77±0.46	98.07±0.53	**98.53±0.26**

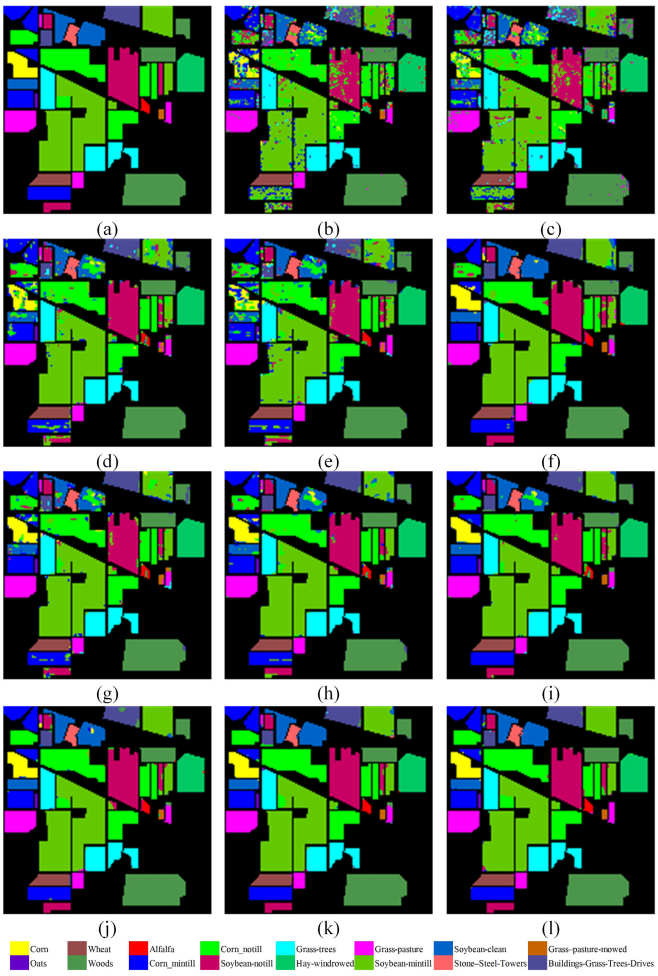


Fig. 6. Classification maps of different algorithms on the IP dataset. (a) Ground-truth map. (b) SVM. (c) RF. (d) ContextNet. (e) RSSAN. (f) SSTN. (g) SSAN. (h) SSSAN. (i) SSAtt. (j) A^2S^2K-ResNet. (k) CVSSN. (l) ANFMG.

it incorporates FODGC module, FODGA module, and FSMF module to effectively reduce the influence of noise and selectively glean detailed information, enhancing OA.

2) Qualitative Analysis: For qualitative evaluation, Figs. 6–13 depict the classification results across various datasets. Specifically, Figs. 6, 8, 10, and 12 exhibit the whole

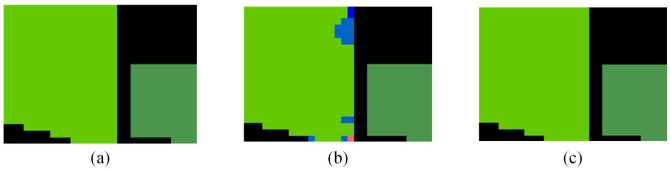


Fig. 7. Suppression effect of the ANFMG algorithm on noise in the IP dataset. (a) Ground-truth map. (b) Portion of the CVSSN classification map. (c) Portion of the ANFMG classification map.

classification maps generated by comparison algorithms and ANFMG, revealing that the ANFMG model attains enhanced accuracy. Furthermore, a subsection of the classification maps, as shown in Figs. 7, 9, 11, and 13, highlight the ability of the ANFMG algorithm to effectively mitigate the impact of noise.

3) Efficiency Analysis: Table VI catalogs the running time of the various methods. As documented in Table VI, among all deep learning models, the ANFMG model ranks third in training time on the IP, KSC, and UH datasets, and fourth on the UP dataset. This ranking is primarily because the ANFMG model invests more time in calculating the Gaussian curvature and mean curvature. Despite this, it is worth highlighting that the training time of the ANFMG model is only one-tenth that of the A^2S^2K-ResNet model, which records the longest training time.

F. Comparison With Spatial Attention Modules

To assess the validity of the proposed FGF-SVS module, we substitute it with prevalent spatial attention modules: SAM from SSAtt, NLB from SSAN, and SSAM from SSSAN. As depicted in Table VII, compared with diverse spatial attention modules, algorithmic performance experiences notable enhancements following the integration of the FGF-SVS module.

G. Analysis of Training and Testing Set Ratios

To further probe the robustness of the 11 different methods under consideration, we analyzed the performance of each method against varying proportions of labeled samples. SVM and RF were excluded from the figure due to the substantial

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TABLE VI

COMPARISON OF MODEL TIME EFFICIENCY OF DIFFERENT METHODS ON DIFFERENT DATASETS

Dataset	Metric	SVM [55]	RF [56]	CVSSN [42]	ContextNet [23]	ANFMG (our)	RSSAN [16]	SSTN [37]	SSAN [15]	SSSAN [36]	SSAtt [14]	A^2S^2K-ResNet [35]
IP	Ttrain(s)	0.05	0.83	57.96	71.78	79.81	80.31	89.89	368.19	316.84	111.32	745.73
	Ttest(s)	1.22	0.17	2.07	2.59	2.96	1.87	2.45	2.40	16.25	3.39	2.54
KSC	Ttrain(s)	0.01	0.32	31.12	33.06	41.13	43.71	46.46	159.49	169.53	56.67	357.21
	Ttest(s)	0.17	0.08	1.66	2.29	2.44	1.93	2.50	2.37	14.15	3.12	2.47
UP	Ttrain(s)	0.21	2.87	244.42	271.29	348.13	340.36	379.60	531.53	1318.45	431.53	2495.13
	Ttest(s)	4.96	0.55	1.37	1.43	2.15	1.82	2.25	2.87	14.66	3.32	2.34
UH	Ttrain(s)	0.05	1.04	83.40	96.73	115.24	122.12	139.01	311.58	466.60	150.77	957.10
	Ttest(s)	1.42	0.21	1.99	2.68	2.82	1.88	2.62	2.33	16.52	3.66	2.49

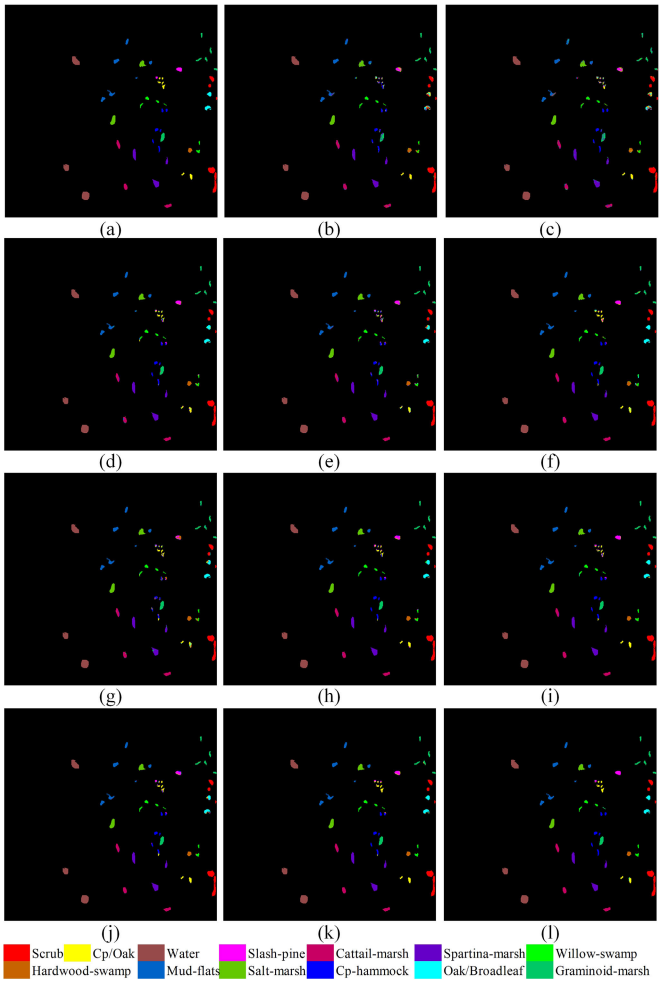


Fig. 8. Classification maps of different algorithms on the KSC dataset. (a) Ground-truth map. (b) SVM. (c) RF. (d) ContextNet. (e) RSSAN. (f) SSTN. (g) SSAN. (h) SSSAN. (i) SSAtt. (j) A^2S^2K-ResNet. (k) CVSSN. (l) ANFMG.

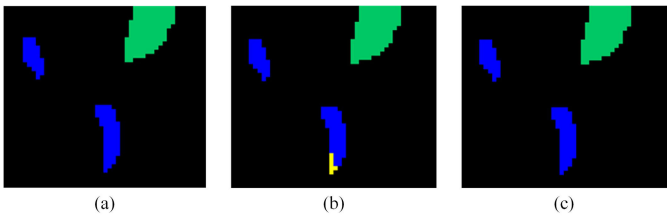


Fig. 9. Suppression effect of the ANFMG algorithm on noise in the KSC dataset. (a) Ground-truth map. (b) Portion of the CVSSN classification map. (c) Portion of the ANFMG classification map.

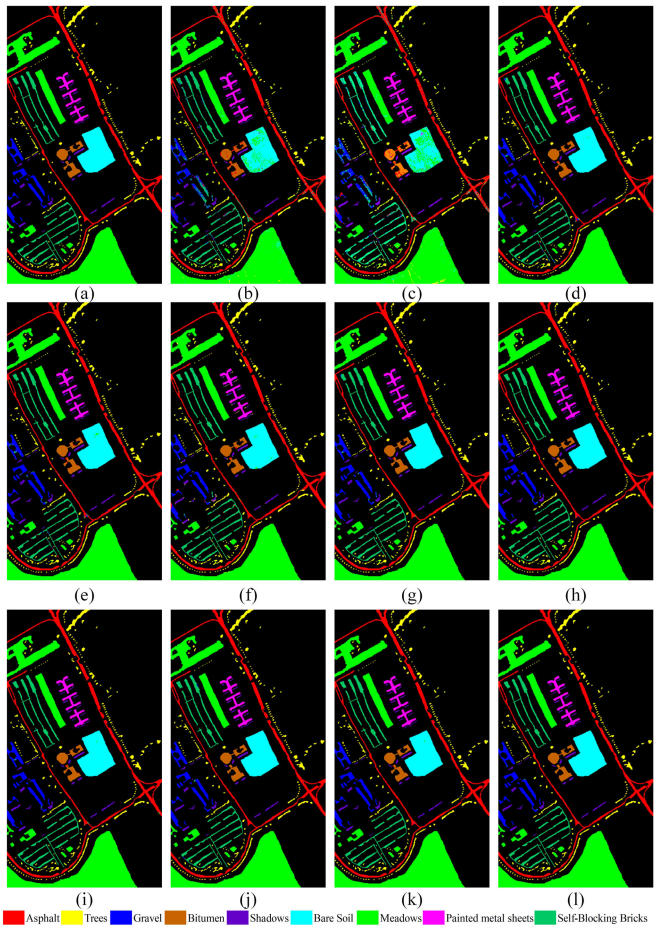


Fig. 10. Classification maps of different algorithms on the UP dataset. (a) Ground-truth map. (b) SVM. (c) RF. (d) ContextNet. (e) RSSAN. (f) SSTN. (g) SSAN. (h) SSSAN. (i) SSAtt. (j) A^2S^2K-ResNet. (k) CVSSN. (l) ANFMG.

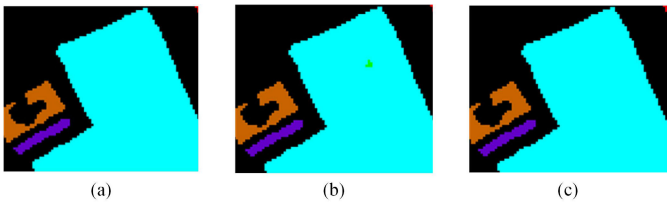


Fig. 11. Suppression effect of the ANFMG algorithm on noise in the UP dataset. (a) Ground-truth map. (b) Portion of the CVSSN classification map. (c) Portion of the ANFMG classification map.

performance gap between them and the other algorithms. Fig. 14 shows that the performance of the nine deep learning methods generally improves with an increasing percentage of training samples, except for the SSTN model. Moreover, the ANFMG model consistently exhibits superior performance across all ratios of training samples. Notably, the enhanced improvements observed in small-sized training sets highlight the model's ability to efficiently extract pivotal features from

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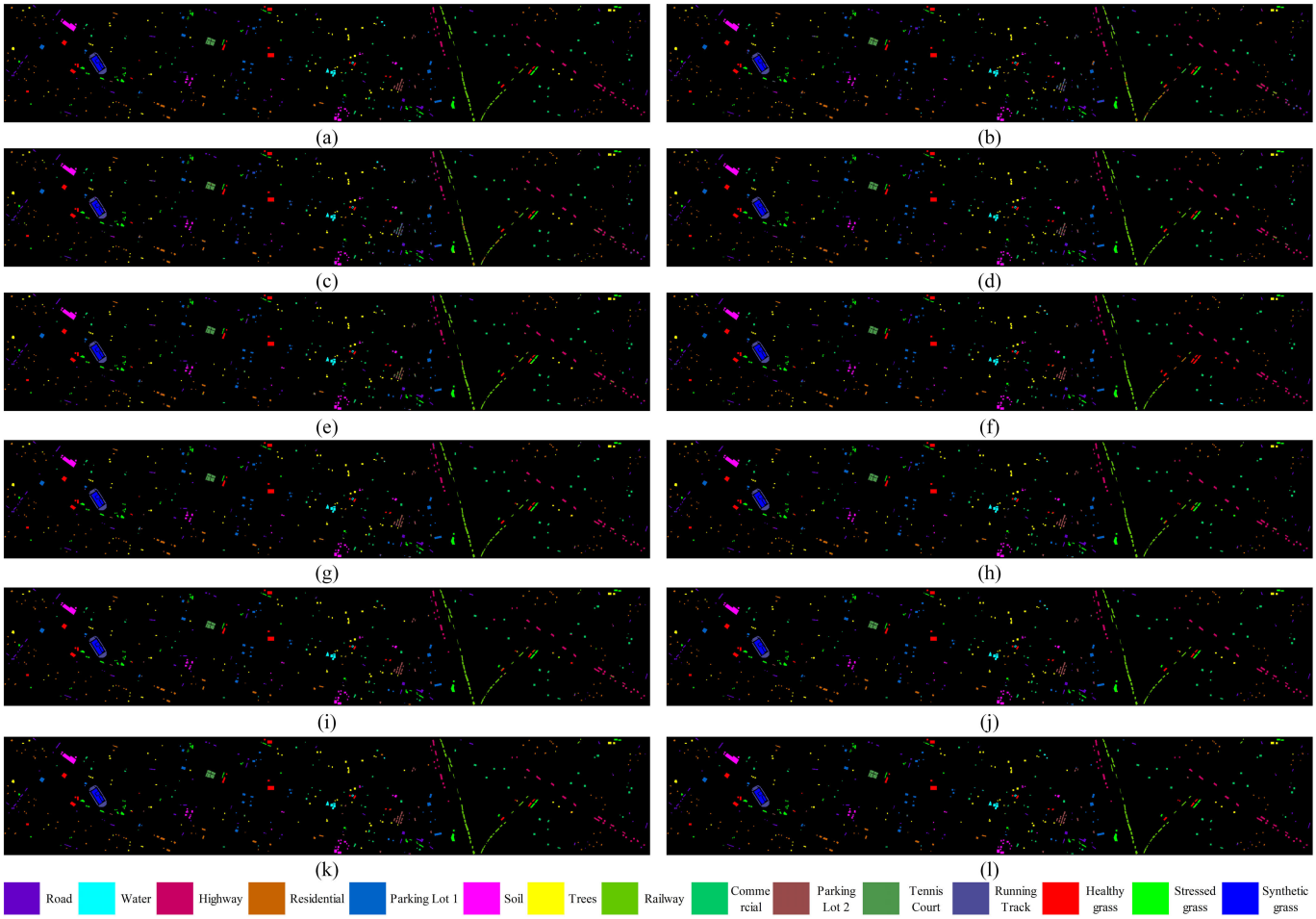


Fig. 12. Classification maps of different algorithms on the UH dataset. (a) Ground-truth map. (b) SVM. (c) RF. (d) ContextNet. (e) RSSAN. (f) SSTN. (g) SSAN. (h) SSSAN. (i) SSAtt. (j) A^{2}S^{2}K-ResNet. (k) CVSSN. (l) ANFMG.

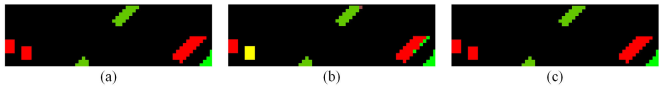


Fig. 13. Suppression effect of the ANFMG algorithm on noise in the UH dataset. (a) Ground-truth map. (b) Portion of the CVSSN classification map. (c) Portion of the ANFMG classification map.

limited data. This further validates the effectiveness of its targeted extraction of detailed features using the fractional-ordered methodology.

TABLE VII
COMPARISON OF SPATIAL ATTENTION MODULES ACROSS DATASETS (%)

Dataset	Indicators	SAM [14]	NLB [15]	SSAM [36]	FGF-SVS (our)
IP	OA (%)	98.10±0.32	98.05±0.24	98.09±0.27	**98.23±0.42**
	AA (%)	96.88±1.32	97.42±0.89	96.67±1.48	**97.65±0.91**
	Kappa (%)	97.84±0.36	97.77±0.27	97.82±0.31	**97.99±0.48**
KSC	OA (%)	98.64±0.29	98.27±0.48	98.64±0.37	**99.00±0.32**
	AA (%)	97.88±0.54	97.12±1.05	97.79±0.66	**98.48±0.56**
	Kappa (%)	98.49±0.33	98.08±0.53	98.49±0.41	**98.89±0.36**
UP	OA (%)	99.89±0.05	99.86±0.05	99.86±0.05	**99.90±0.05**
	AA (%)	99.85±0.07	99.82±0.10	99.83±0.08	**99.86±0.09**
	Kappa (%)	99.85±0.07	99.82±0.07	99.82±0.07	**99.87±0.06**
UH	OA (%)	98.46±0.41	98.44±0.37	98.46±0.37	**98.64±0.24**
	AA (%)	98.48±0.34	98.50±0.37	98.54±0.34	**98.61±0.33**
	Kappa (%)	98.34±0.45	98.31±0.40	98.34±0.40	**98.53±0.26**

H. Classification Results With Spatially Disjoint Samples

To validate the classification outcomes of this algorithm, we utilize the second partitioning method to split the UP, UH, and LK datasets into distinct training and testing sets and undertake comparative and ablation experiments.

1) Comparative Experiment: In this experiment, the proposed ANFMG is compared against ten related algorithms. The experimental findings are tabulated in Table VIII. However, because function A^{2}S^{2}K-ResNet has high memory requirements in the LK dataset, it could not run in our experimental environment, resulting in missing data in Table VIII. Notably, compared to previous classification results, nearly all methods exhibit a substantial performance decline. This decline is attributed to the fact that randomly selecting samples overlooks the spatial dependence between training and testing sets, leading to an overestimation of classification performance. Despite experiencing a performance decline, ANFMG consistently attains the highest OA, AA, and Kappa across the UP, UH, and LK datasets, with its decline being the most minimal on the UP dataset. Therefore, even when using spatially disjoint samples, the classification performance of ANFMG remains superior to that of other algorithms. Besides, Table VIII shows relatively low standard deviations, indicating good robustness of ANFMG.

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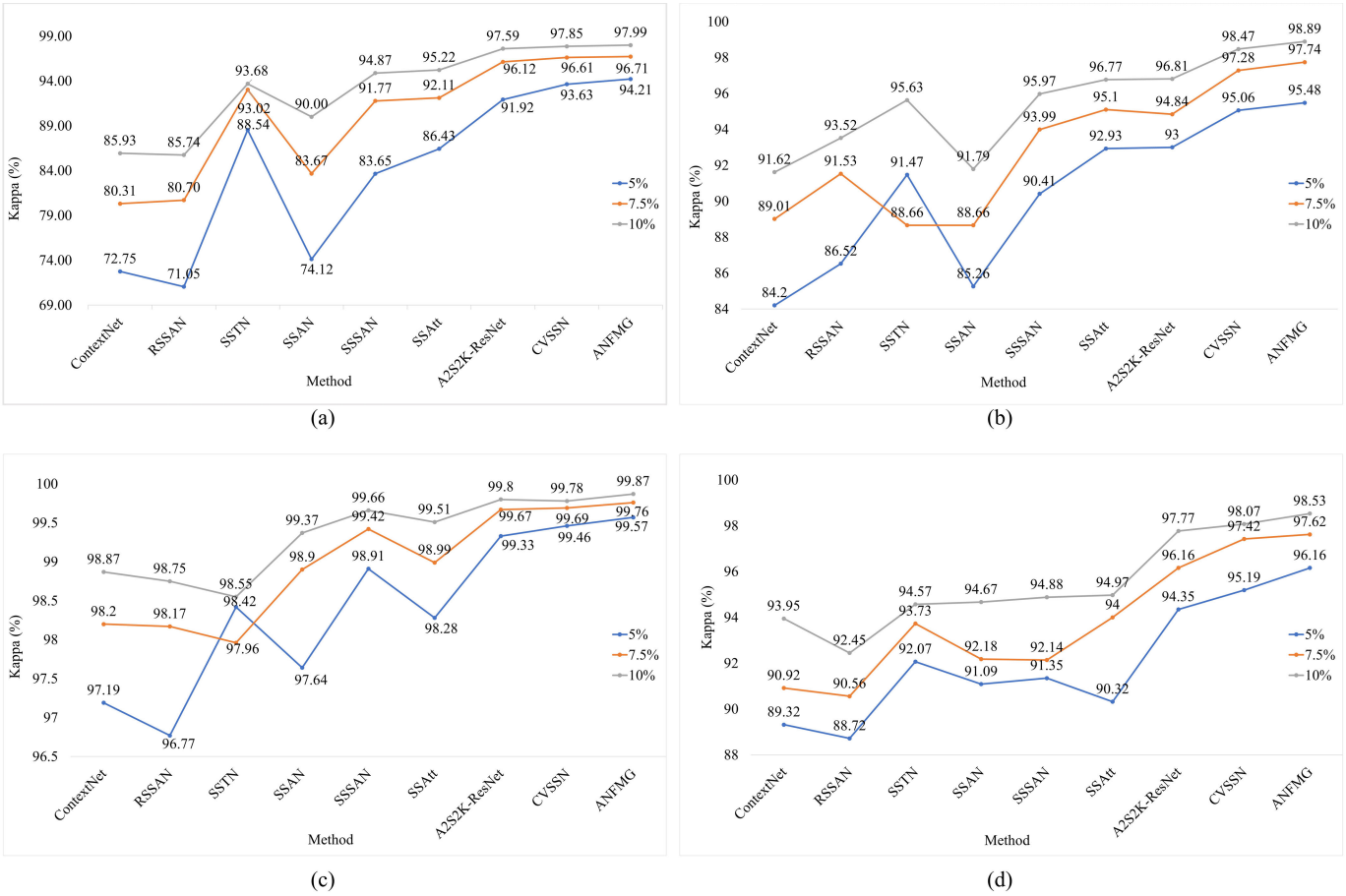


Fig. 14. Comparison of Kappa (%) values for various methods across different datasets with 5%, 7.5%, and 10% training sample sizes. (a) IP dataset. (b) KSC dataset. (c) UP dataset. (d) UH dataset.

TABLE VIII

CLASSIFICATION EFFECTS OF DIFFERENT ALGORITHMS UNDER VARIOUS DATASETS WITH SPATIALLY DISJOINT SAMPLES (%)

Dataset	Indicators	SVM [55]	RF [56]	ContextNet [23]	RSSAN [16]	SSTN [37]	SSAN [15]	SSSAN [36]	SSAtt [14]	A^2S^2K-ResNet [35]	CVSSN [42]	ANFMG (our)
UP	OA (%)	75.86±0.01	70.37±0.01	85.49±0.77	85.91±2.00	87.64±2.42	88.17±0.90	87.61±1.47	89.47±0.47	85.49±3.22	91.07±0.64	**91.93±1.05**
	AA (%)	76.58±0.02	72.20±0.02	83.35±0.96	83.28±2.95	87.70±2.95	86.44±1.32	88.15±2.00	89.72±0.59	88.36±1.81	91.89±1.04	**92.42±1.06**
	Kappa (%)	69.76±0.02	63.00±0.02	80.54±1.01	80.97±2.58	83.62±3.19	84.00±1.21	83.34±1.84	85.66±0.65	81.26±3.85	87.85±0.84	**89.01±1.45**
LK	OA (%)	91.23±0.00	85.96±0.00	96.35±0.31	96.98±0.65	98.09±0.74	97.27±0.21	98.37±0.36	97.95±0.16		99.30±0.04	**99.36±0.08**
	AA (%)	78.08±0.00	70.35±0.00	88.48±1.00	89.70±1.82	93.51±2.02	90.45±0.71	93.97±1.07	92.42±0.49		97.61±0.14	**97.75±0.27**
	Kappa (%)	88.72±0.00	82.16±0.00	95.24±0.40	96.06±0.84	97.50±0.97	96.43±0.28	97.86±0.47	97.32±0.20		99.08±0.05	**99.16±0.10**
UH	OA (%)	76.76±0.02	72.91±0.02	74.70±1.08	76.11±1.04	79.25±1.51	77.24±1.05	81.07±1.01	80.28±0.69	82.58±0.87	82.22±1.05	**82.81±0.63**
	AA (%)	78.49±0.02	75.54±0.02	77.07±1.29	77.60±1.30	80.68±2.25	79.93±1.27	83.67±1.56	82.75±0.83	85.90±1.70	85.15±1.40	**85.98±0.92**
	Kappa (%)	74.98±0.02	70.90±0.02	72.65±1.16	74.19±1.13	77.61±1.63	75.39±1.12	79.55±1.09	78.68±0.75	81.22±0.95	80.78±1.14	**81.43±0.67**

TABLE IX

COMPARISON OF CLASSIFICATION RESULTS OF ANFMG AND ITS VARIANTS ACROSS DATASETS WITH SPATIALLY DISJOINT SAMPLES (%)

Dataset	Indicators	ANFMG-FC	ANFMG-F	ANFMG
UP	OA (%)	91.07±0.64	91.31±0.95	91.93±1.05
	AA (%)	91.89±1.04	92.38±0.92	92.42±1.06
	Kappa (%)	87.85±0.84	88.16±1.29	89.01±1.45
LK	OA (%)	99.30±0.04	99.32±0.07	99.36±0.08
	AA (%)	97.61±0.14	97.71±0.23	97.75±0.27
	Kappa (%)	99.08±0.05	99.11±0.09	99.16±0.10
UH	OA (%)	82.22±1.05	82.46±0.70	82.81±0.63
	AA (%)	85.15±1.40	85.74±0.93	85.98±0.92
	Kappa (%)	80.78±1.14	81.02±0.76	81.43±0.67

2) Ablation Experiment: To delve deeper into the proposed differential geometric measurement and FOD contributions, we performed ablation experiments. The results are illustrated in Table IX.

Compared to ANFMG, after the removal of FOD, ANFMG-F shows varying degrees of decline in OA, AA, and Kappa coefficients across the datasets. Likewise, Compared to ANFMG-F, after the removal of differential geometric measurement, ANFMG-FC also experienced a decline in OA, AA, and Kappa across the datasets. Therefore, both differential geometric measurement and FOD are proven effective.

V. CONCLUSION

This article proposes an ANFMG features for classifying HSI. First, we present the specific fractal detail extraction strategy, designed for the selective extraction of detailed information, enabling efficient identification of useful features in small-scale datasets. Second, we construct FODGC and FODGA modules to flexibly suppress noise. Finally, we design the FSMF module, which adaptively adjusts the contribution of central vectors and their adjacent vectors to the algorithm,

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addressing classification errors stemming from central pixels that exhibit spectral variability. Experimental results validate that this proposed model outperforms prevailing models in terms of OA, AA, and Kappa. Future research will focus on exploring a more streamlined approach for computing curvature, to minimize the ANFMG model's training duration.

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